

1, AJALOMI OGBORU COPMOUND,
ABRAKA, DELTA STATE,
NIGERIA.

17th May 2025

The SE Cai Selection Committee,

TU Dresden

01069 Dresden,

Germany.

Dear Members of the Selection Committee,

LETTER OF MOTIVATION IN VIEW STUDY FOR MASTERS IN ARTIFICIAL INTELLIGENCE

I am writing to express my sincere interest in the SE Cai Scholarship for master's studies in Artificial Intelligence. I have a degree in Mechanical Engineering with strong passion for applying AI to energy systems and predictive maintenance applications, I am eager to pursue a Master's degree aligned with SE Cai's current research goals. I am therefore applying to the M.Sc. program in Computational Modeling and Simulation at TU Dresden, specifically in the Applied Artificial Intelligence track.

During my undergraduate studies, I was awarded a scholarship by Midwestern oil and gas because of my academic performance which enabled me to successfully complete my undergraduate course on mechanical engineering. Alongside my studies I also completed various short courses such as Elements of Artificial Intelligence from the University of Helsinki, Introduction to Machine Learning, and an internship program on Artificial Intelligence from Pantech University, from which certifications were obtained for these programs. With my passion for AI and experiences from these programs, I have been in the business for using various machine learning and AI techniques for various engineering applications such as the development of predictive maintenance tools and various engineering systems in which AI and machine learning models were implemented. For my final thesis, I designed and fabricated a solar-powered cold room system integrated with a pump thermal energy storage system for post-harvest tomato storage, incorporating energy-efficient principles and control mechanisms. This project sparked further deep interest in integrating intelligent systems into sustainable technologies. I later extended my research to include exergy analysis using TESP and Aspen Plus for optimizing refrigeration systems—work that strongly aligns with AI-based modeling and simulation techniques promoted by SE Cai.

I've been actively involved in academic mentorship and skill-sharing initiatives, including teaching SolidWorks, Python app development, and machine learning workshops tailored to real-world applications in energy sustainability and agriculture. I am particularly passionate about democratizing

access to technology, and my future goals include contributing to more environmental friendly innovations using AI for developing regions.

My academic record, reflected in a strong performance in thermodynamics, energy systems, and python programming, meets the grade requirement after conversion. More importantly, my commitment to AI for societal benefit and energy sustainability resonates with SE Cai's mission. As a candidate from a developing country, this scholarship would not only enable me to focus fully on my academic journey but also help me build impactful research collaborations.

Thank you for considering my application. I look forward to the opportunity to join TU Dresden's vibrant academic community and contribute meaningfully to the SE Cai initiative.

Yours sincerely,

OGBORU REWARD OGHENEFEJIRO

E-mail: rewardfejiro777@gmail.com